

Assignment

Royal University of Phnom Penh

Department of Computer Science

Major: Window Server Admin

Professor: Assoc.Prof Dr.Ouk Khean

Email address: khean_ouk@yahoo.com.sg/oukkhean@gmail.com

Group: 2

Class: A2_Y4

Homework: 8

Group Members

1. ជា គីមឡេង
2. ហេង ឡាយ
3. ម៉ឿន មករា
4. ផាន ដាវីត
5. ស្រួយ រតនា

Homework 8

1: Install Apache and OpenSSL

```

root@localhost:~
[henglay@localhost ~]$ su -
Password:
[root@localhost ~]# sudo yum install httpd mod_ssl openssl -y
sudo systemctl start httpd
sudo systemctl enable httpd
CentOS Stream 9 - BaseOS                3.2 kB/s | 4.2 kB    00:01
CentOS Stream 9 - BaseOS                516 kB/s | 8.9 MB    00:17
CentOS Stream 9 - AppStream             3.4 kB/s | 4.2 kB    00:01
CentOS Stream 9 - AppStream            106 kB/s | 26 MB     04:16
CentOS Stream 9 - Extras packages       3.0 kB/s | 5.6 kB    00:01
Package httpd-2.4.62-10.el9.x86_64 is already installed.
Package openssl-1:3.5.1-6.el9.x86_64 is already installed.
Dependencies resolved.

=====
Package                        Architecture      Version           Size              Repository
=====
Installing:
mod_ssl                        x86_64            1:2.4.62-10.el9   110 k             appstream

Transaction Summary
=====
Install 1 Package

Total download size: 110 k
Installed size: 273 k
Downloading Packages:
mod_ssl-2.4.62-10.el9.x86_64.rpm        140 kB/s | 110 kB    00:00
-----
Total                                58 kB/s | 110 kB    00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing    : mod_ssl-1:2.4.62-10.el9.x86_64 1/1
  Running scriptlet: mod_ssl-1:2.4.62-10.el9.x86_64 1/1
  Verifying     : mod_ssl-1:2.4.62-10.el9.x86_64 1/1

Installed:
  mod_ssl-1:2.4.62-10.el9.x86_64

Complete!
[root@localhost ~]#

```

2: Generate the RSA SSL Certificate

[illegible]

3: Create the Website Interface

```
[root@localhost ~]# sudo mkdir -p /var/www/group2A2y4/public_html
sudo vi /var/www/group2A2y4/public_html/index.html
```

```

    <span class="tag">UI/UX Designer</span>
    <div class="divider"></div>
    <p>Creating intuitive interfaces with a focus on user behavior.</p>
  </div>
</div>

<div class="prof-card">
  <div class="image-wrapper">
    
    <div class="status-dot"></div>
  </div>
  <div class="card-info">
    <h3>Chea Kim Leng</h3>
    <span class="tag">UI/UX Designer</span>
    <div class="divider"></div>
    <p>Creating intuitive interfaces with a focus on user behavior.</p>
  </div>
</div>

<div class="prof-card">
  <div class="image-wrapper">
    
    <div class="status-dot"></div>
  </div>
  <div class="card-info">
    <h3>Meoun Makara</h3>
    <span class="tag">UI/UX Designer</span>
    <div class="divider"></div>
    <p>Creating intuitive interfaces with a focus on user behavior.</p>
  </div>
</div>

<div class="prof-card">
  <div class="image-wrapper">
    
    <div class="status-dot"></div>
  </div>
  <div class="card-info">
    <h3>Sroy Rathana</h3>
    <span class="tag">UI/UX Designer</span>
    <div class="divider"></div>
    <p>Creating intuitive interfaces with a focus on user behavior.</p>
  </div>
</div>

```

4: Configure Virtual Host for SSL

```
[root@localhost ~]# sudo vi /etc/httpd/conf.d/group2A2y4.conf
[root@localhost ~]# sudo vi /etc/httpd/conf.d/group2A2y4.conf
[root@localhost ~]# sudo systemctl restart httpd
```

```
root@localhost:-
```

```
<VirtualHost *:443>  
    ServerName www.group2A2y4.edu.kh  
    DocumentRoot /var/www/group2A2y4/public_html  
  
    SSLEngine on  
    SSLCertificateFile /etc/httpd/ssl/apache.crt  
    SSLCertificateKeyFile /etc/httpd/ssl/apache.key  
  
    <Directory /var/www/group2A2y4/public_html>  
        AllowOverride All  
        Require all granted  
    </Directory>  
</VirtualHost>
```

```
" /etc/httpd/conf.d/group2A2y4.conf" 13L, 360B
```

```
All
```

5: Map the Local Domain

```
root@localhost:~  
126.0.0.1 localhost localhost.localdomain localhost4 localhost4.localdomain4  
127.0.0.1 www.group2A2y4.edu.kh:::1 localhost localhost.localdomain localhost6 localhost6.localdomain6  
127.0.0.1 www.group2A2y4.edu.kh
```

6: Firewall and Testing

```
[root@localhost ~]# sudo firewall-cmd --permanent --add-service=https
sudo firewall-cmd --reload
success
success
[root@localhost ~]# sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-01-29 14:36:31 +07; 8min ago
     Docs: man:httpd.service(8)
  Main PID: 3514 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
     Tasks: 177 (limit: 22773)
    Memory: 15.0M (peak: 15.5M)
       CPU: 642ms
    CGroup: /system.slice/httpd.service
            └─3514 /usr/sbin/httpd -DFOREGROUND
              └─3515 /usr/sbin/httpd -DFOREGROUND
                └─3516 /usr/sbin/httpd -DFOREGROUND
                  └─3518 /usr/sbin/httpd -DFOREGROUND
                    └─3524 /usr/sbin/httpd -DFOREGROUND

Jan 29 14:36:31 localhost.localdomain systemd[1]: Starting The Apache HTTP Server...
Jan 29 14:36:31 localhost.localdomain httpd[3514]: Server configured, listening on: port 443, port 80
Jan 29 14:36:31 localhost.localdomain systemd[1]: Started The Apache HTTP Server.
[root@localhost ~]# ping www.group2A2y4.edu.kh
ping: www.group2A2y4.edu.kh: Name or service not known
[root@localhost ~]# ifconfig
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.17.128 netmask 255.255.255.0 broadcast 192.168.17.255
    inet6 fe80::20c:29ff:fe72:5090 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:72:50:90 txqueuelen 1000 (Ethernet)
    RX packets 145552 bytes 209062364 (199.3 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 41662 bytes 3002014 (2.8 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 17 bytes 2062 (2.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 17 bytes 2062 (2.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

7.RSA

Issuer Name	
Country	ca
State/Province	168
Locality	pp
Organization	rupp
Organizational Unit	rupp
Common Name	www.group2A2y4.edu.kh
Email Address	lay123@gmail.com
Validity	
Not Before	Thu, 29 Jan 2026 07:33:48 GMT
Not After	Fri, 29 Jan 2027 07:33:48 GMT
Public Key Info	
Algorithm	RSA
Key Size	2048
Exponent	65537
Modulus	9F:80:F8:27:73:BB:D1:17:3F:41:8A:14:13:F4:A1:C6:CF:D6:04:2D:B4:B8:32:4E:...
Miscellaneous	
Serial Number	52:66:6F:C4:6B:56:5D:B7:5E:AD:97:C5:58:36:DF:5E:39:1D:FE:3B
Signature Algorithm	SHA-256 with RSA Encryption
Version	3
Download	PEM (cert) PEM (chain)